# Azure Application Gateway SSL Certificate Expiry Notification

*Azure Automation Runbook-Based SSL Certificate Health Monitoring*

DEVELOPED BY THE DEVOPS TEAM @ MOVATE

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| --- | --- | --- |
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# Table of Contents


1. Executive Summary
2. Business Problem
3. Solution Overview
4. High-Level Architecture (HLD)
5. Low-Level Design (LLD)
6. Key Technical Highlights
7. Certificate Status Classification
8. Technology Stack
9. Code Structure
10. Configuration
11. Deployment
12. IAM & Security Model
13. Email & Excel Output
14. Advantages
15. Limitations
16. Use Cases
17. Operational Runbook
18. Final Outcome

## 1 Executive Summary

This solution is an **Azure Automation Python 3 Runbook** that proactively monitors SSL/TLS certificate expiry across all Azure Application Gateways in one or more subscriptions. It runs on a daily schedule, scans every certificate attached to a live TLS listener, and delivers a colour-coded health report — an HTML email plus a full Excel attachment — via **Azure Communication Services (ACS) Email**.

The runbook is entirely **read-only**: it never creates, modifies, or deletes any Azure resource.

### Key Outcomes

- Proactive
- Multi-Subscription
- Read-Only
- Auditable

| Outcome | Description |
|---|---|
| **Proactive** | Catches certificates before they expire — not after outages occur |
| **Multi-Subscription** | Single runbook covers all Application Gateways across the entire Azure estate |
| **Read-Only** | No risk of accidental resource modification; safe to run with Reader-only RBAC |
| **Auditable** | Full Excel report delivered after every run for compliance and governance records |

> **Zero-touch monitoring:** once deployed, the runbook runs daily on a schedule with no manual intervention required.

## 2 Business Problem

| Challenge | Impact |
|---|---|
| Manual certificate tracking | Human error and missed expiry deadlines across multiple gateways |
| No centralised visibility | Poor governance across multiple subscriptions and Application Gateways |
| Expired certificates | Application outages, browser security warnings, broken HTTPS endpoints |
| Reactive monitoring | Certificates discovered as expired only after end-user impact has occurred |

## 3 Solution Overview

An **Azure Automation Python 3 Runbook** executes on a daily schedule to perform the full certificate health check automatically:

1. **Authenticate** to Azure using a Service Principal ( `ClientSecretCredential` )
2. **Discover** every Application Gateway across all configured subscriptions
3. **Fetch** full gateway details including `publicCertData` for each SSL certificate
4. **Parse** certificate expiry dates from PKCS#7/DER/PEM encoded certificate data
5. **Classify** each certificate: Expired / Critical / Warning / OK / Unknown
6. **Report** results via a colour-coded HTML email and a full Excel attachment

### Certificate Status Thresholds

| Status | Condition |
|---|---|
| **Expired** | Days remaining < 0 — certificate has already expired |
| **Critical** | 0 – 7 days remaining — requires immediate action |
| **Warning** | 8 – 14 days remaining (configurable via `AG_ALERT_DAYS` ) |
| **OK** | > 14 days remaining — certificate is healthy |
| **Unknown** | Expiry could not be parsed (e.g. Key Vault-backed certificates) |

# 4 High-Level Architecture (HLD)

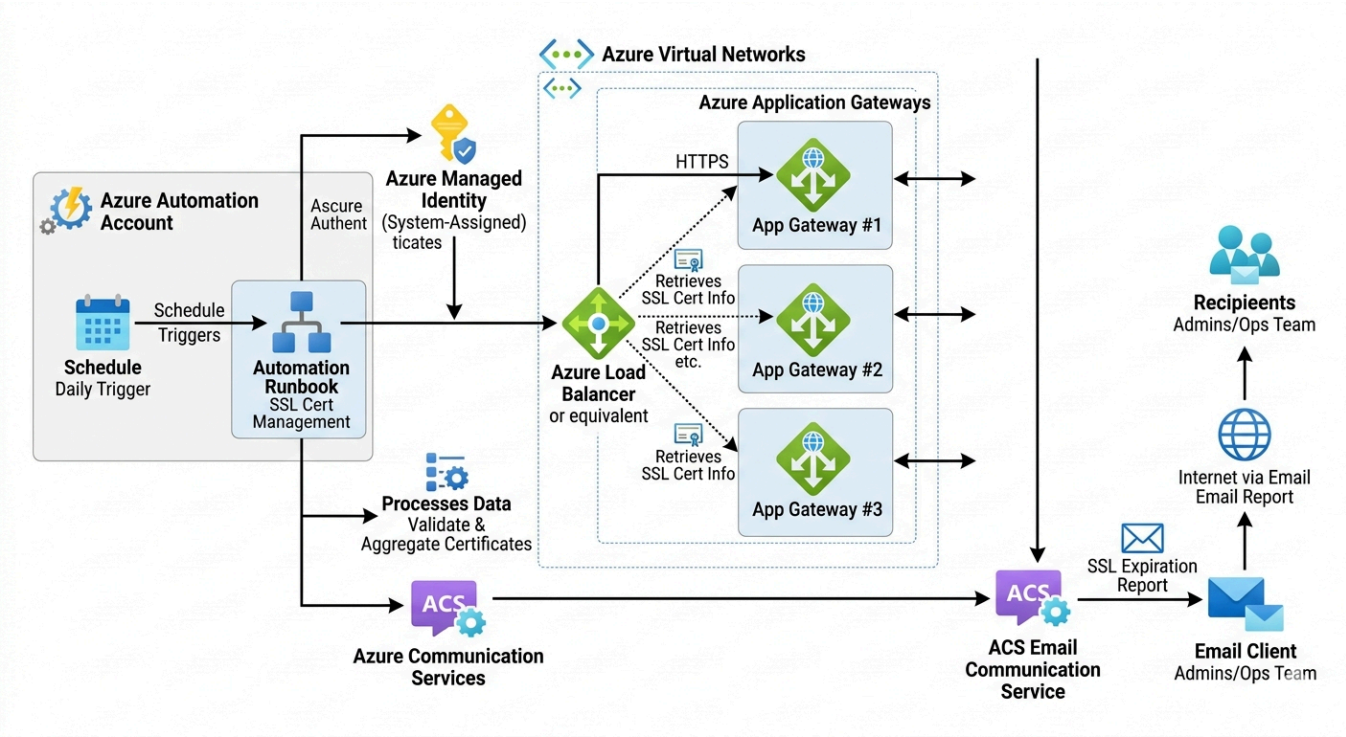



*Figure 1 — Sample HTML Email Report: colour-coded status table with metrics summary card*

## Components

| Component | Role |
|---|---|
| **Azure Automation Account** | Hosts and schedules the Python 3 runbook |
| **Azure Automation Schedule** | Triggers the runbook daily (e.g. 06:00 UTC) |
| **Service Principal** | Identity used by the runbook to read Application Gateway resources (Reader role) |
| **Application Gateways** | Target resources — SSL certificate data is read from each gateway via Azure API |
| **Azure Communication Services** | Delivers the HTML email and Excel attachment to the recipient list |

### Architecture Diagram

```
┌──────────────────────────────────────────────────────────────────────────┐
│                              Azure Subscription(s)                       │
│                                                                          │
│  ┌──────────────────────────┐    ┌──────────────────────────────────┐   │
│  │  Azure Automation Account │    │      Application Gateways        │   │
│  │  ┌────────────────────┐   │    │  ┌──────────┐  ┌──────────────┐ │   │
│  │  │  Python 3 Runbook  │──┼────┼──►│ Gateway 1 │  │   Gateway N  │ │   │
│  │  │  (main.py)         │   │    │  │ SSL Certs │  │   SSL Certs  │ │   │
│  │  └────────────────────┘   │    │  └──────────┘  └──────────────┘ │   │
│  │  ┌────────────────────┐   │    └──────────────────────────────────┘   │
│  │  │  Schedule (daily)  │   │                                          │
│  │  └────────────────────┘   │    ┌──────────────────────────────────┐   │
│  │  ┌────────────────────┐   │    │  Azure Communication Services (ACS) │   │
│  │  │  Variables / Creds │   │    │  Email                           │   │
│  │  │  - AzureSPCredential│──┼────┼►  Send HTML + Excel to recipients │   │
│  │  │  - ACS_CONNECTION.. │   │    └──────────────────────────────────┘   │
│  │  └────────────────────┘   │                                          │
│  └──────────────────────────┘                                          │
└──────────────────────────────────────────────────────────────────────────┘

Authentication Flow:
  Automation Account → ClientSecretCredential (Service Principal)
                     → Azure RBAC: Reader on each subscription
```

## 5 Low-Level Design (LLD)

### Execution Flow

Trigger — Azure Automation Schedule (daily)

↓

Phase 1 — Authenticate via Service Principal

↓

Phase 2 — Discover & Scan Application Gateways

↓

Phase 3 — Parse & Classify Certificates

↓

Phase 4 — Build Excel Report & HTML Email

↓

Phase 5 — Send via Azure Communication Services

### Detailed Phase Breakdown

| Phase | Actions |
|---|---|
| **Authenticate** | Reads `AzureSPCredential` from Automation Account; builds `ClientSecretCredential`; instantiates `NetworkManagementClient` per subscription |
| **Discover & Scan** | Calls `list_all()` to enumerate all Application Gateways; calls `get()` per gateway for full `publicCertData`; filters to certificates attached to active TLS listeners |
| **Parse & Classify** | Attempts 4-stage fallback decode: PKCS#7 → bare DER → PEM-in-base64 → raw PEM; classifies each cert by days remaining; marks Unknown if expiry cannot be parsed |
| **Build Reports** | Builds colour-coded Excel workbook sorted by days remaining; builds HTML email with metrics card and per-status top-5 tables |
| **Send** | Sends HTML email + Excel attachment via ACS Email; `DRY_RUN=true` writes to disk for local testing |

> **4-stage certificate parsing:** Azure Application Gateways return certificate data in multiple formats depending on source and version. The 4-stage fallback chain (PKCS#7 → DER → PEM-in-base64 → raw PEM) ensures maximum compatibility across all gateway configurations.

## 6 Key Technical Highlights

### Resilient Certificate Parsing

- Four-stage fallback decode chain handles every certificate format Azure may return
- PKCS#7 container → bare DER → PEM-in-base64 → raw PEM string
- Unknown status clearly flagged for Key Vault-backed certificates where `publicCertData` is not exposed

### Listener-Aware Scanning

- Only certificates attached to active TLS listeners are reported
- Orphaned / unused certificates are intentionally skipped to reduce noise in reports

### Read-Only Safety

- Zero write operations against any Azure resource
- Safe to run with minimal RBAC permissions — `Reader` role only
- `DRY_RUN=true` mode writes output to disk without sending any email

### Dual Output Format

- **HTML email** — at-a-glance metrics card with colour-coded per-status tables
- **Excel attachment** — full 13-column report with auto-filter and row-level colour coding

### Configuration-Driven

- All parameters stored in Automation Account variables — no redeployment needed to update config
- Warning threshold (`AG_ALERT_DAYS`), subscription list, and recipient list are all runtime-configurable

## 7 Certificate Status Classification

| Status | Days Remaining | Row Colour | Description |
|---|---|---|---|
| Expired | < 0 | Dark Red | Certificate has already expired — immediate replacement required |
| Critical | 0 – 7 | Orange-Red | Expiry imminent — escalate and act immediately |
| Warning | 8 – 14 | Amber | Action required within the warning window (configurable) |
| OK | > 14 | Green | Certificate is healthy — no action required |
| Unknown | N/A | Grey | Expiry cannot be parsed — typically Key Vault-backed certificates |

> The Warning threshold is configurable via the `AG_ALERT_DAYS` Automation variable (default: `14`). Adjust this value to match your organisation's certificate renewal lead time without redeploying the runbook.

## 8 Technology Stack

| Layer | Technology |
|---|---|
| Compute | Azure Automation Account (Python 3 Runbook) |
| Scheduler | Azure Automation Schedule (daily trigger) |
| Authentication | Azure Service Principal — `ClientSecretCredential` from `azure-identity` |
| Resource Discovery | Azure Management SDK — `azure-mgmt-network`, `azure-mgmt-subscription` |
| Certificate Parsing | Python `cryptography` library — PKCS#7, X.509, DER, PEM |
| Email | Azure Communication Services (ACS) Email — `azure-communication-email` |
| Reporting | `openpyxl` (Excel .xlsx), HTML (inline email template) |
| Language | Python 3.8 (Azure Automation runtime) |

### Python Dependencies

| Package | Purpose |
|---|---|
| `azure-identity` | Service Principal authentication |
| `azure-mgmt-network` | Application Gateway list/get API |
| `azure-mgmt-subscription` | Resolve subscription display names |
| `azure-communication-email` | Send email via ACS |
| `cryptography` | Parse PKCS#7 / X.509 certificate data |
| `openpyxl` | Build Excel .xlsx workbook |
| `python-dotenv` | Load .env for local runs (not required in Automation Account) |

## 9 Code Structure

```
ssl-certificate-notifier/
├── main.py                    # Runbook / main script (all logic in one file)
├── requirements.txt           # Python dependencies
├── DOCUMENTATION.md           # Full technical documentation
└── .env                       # Local secrets (never commit — git-ignored)
```

### main.py — Section Breakdown

| Section | Responsibility |
|---|---|
| **Section 1 — Configuration** | Resolves all settings from Automation variables, env vars, or defaults. `DRY_RUN=true` writes outputs to disk without sending email. |
| **Section 2 — Certificate Scanning** | Read-only Azure API calls. Lists/gets Application Gateways, parses cert data using 4-stage fallback, builds flat records per certificate. |
| **Section 3 — Excel Report** | Builds colour-coded `.xlsx` workbook using `openpyxl`. Rows sorted by days remaining (soonest first). |
| **Section 4 — HTML Email** | Builds responsive HTML email with metrics card and per-status tables (Expired / Critical / Warning / Unknown). |
| **Section 5 — ACS Email** | Sends the HTML email body + Excel attachment via `azure-communication-email`. |
| **Section 6 — Entry Point** | `main()` orchestrates sections 1–5 in sequence. |

## 10 Configuration

Config resolution order for every setting:

1. Azure Automation variable / credential asset
2. Environment variable (local `.env` via `python-dotenv`)
3. Hard-coded default, or `sys.exit(1)` for required values

### Azure Automation Account Assets

| Asset Type | Name | Description |
|---|---|---|
| Credential | `AzureSPCredential` | Username = Client ID, Password = Client Secret |
| Variable (String) | `AZURE_TENANT_ID` | Azure AD tenant ID |
| Variable (String) | `AG_SUBSCRIPTION_IDS` | Comma-separated subscription IDs to scan |
| Variable (String) | `AG_ALERT_DAYS` | Warning threshold in days before expiry (default: `14`) |
| Variable (String) | `ACS_CONNECTION_STRING` | Azure Communication Services connection string |
| Variable (String) | `ACS_SENDER_ADDRESS` | Verified ACS sender email address |
| Variable (String) | `AG_NOTIFICATION_TO` | Comma-separated recipient email addresses |

### Local Development (.env file)

```
AZURE_CLIENT_ID=<service-principal-client-id>
AZURE_CLIENT_SECRET=<service-principal-client-secret>
AZURE_TENANT_ID=<tenant-id>
AG_SUBSCRIPTION_IDS=<sub-id-1>,<sub-id-2>
AG_ALERT_DAYS=14
DRY_RUN=false
ACS_CONNECTION_STRING=<acs-connection-string>
ACS_SENDER_ADDRESS=DoNotReply@<domain>.azurecomm.net
AG_NOTIFICATION_TO=team@example.com
```

> Set `DRY_RUN=true` to skip the ACS call and write the Excel and HTML files to disk locally for inspection. No Azure Email resources are consumed in dry-run mode.

# 11 Deployment

## Prerequisites

1. An Azure Automation Account with Python 3 runbook support
2. A Service Principal with `Reader` role on all target subscriptions
3. An Azure Communication Services resource with a verified sender domain
4. Required Python packages installed in the Automation Account

## Step-by-Step Deployment

| Step | Action |
|---|---|
| **1. Install packages** | Upload `azure-identity`, `azure-mgmt-network`, `azure-mgmt-subscription`, `azure-communication-email`, `cryptography`, `openpyxl` to the Automation Account Python 3 packages |
| **2. Import runbook** | Upload `main.py` as a Python 3 runbook in the Automation Account |
| **3. Configure assets** | Add all variables and credentials listed in the Configuration section |
| **4. Create schedule** | Link a daily schedule to the runbook (recommended: 06:00 UTC) |
| **5. Test run** | Trigger the runbook manually from the Azure Portal to validate end-to-end connectivity |

## Local Test Run

```
pip install -r requirements.txt
DRY_RUN=true python main.py       # writes .xlsx + .html to disk, no email sent
```

# 12 IAM & Security Model

## Authentication Layers

| Layer | Identity | Permissions |
|---|---|---|
| **Automation Account** | Managed Identity / Run As | Access to Automation variables and credentials |
| **Application Gateway scanning** | Service Principal (`AzureSPCredential`) | `Reader` role on each target subscription |
| **Email delivery** | ACS connection string | Authenticated ACS Email sender permission |

### Required RBAC

The Service Principal requires **Reader** role on each target subscription (or at minimum on resource groups containing Application Gateways). No write permissions are required — the runbook is entirely read-only.

### Security Guidelines

> **These rules apply to every deployment of this runbook:**
>
> - Store all secrets in **Azure Automation encrypted variables / credentials** — never hardcode in the runbook source
> - The `.env` file is for local development only — add it to `.gitignore` and never commit to source control
> - Scope the Service Principal to the **minimum required subscriptions** with `Reader` role only
> - Rotate the Service Principal client secret on your organisation's standard credential rotation schedule

## 13 Email & Excel Output

### HTML Email Structure

| Section | Description |
|---|---|
| **Metrics card** | "At a glance" summary — total certs scanned, alert badge counts per status, report timestamp |
| **Expired table** | Top-5 expired certificates (dark red theme) |
| **Critical table** | Top-5 certificates expiring within 7 days (orange-red theme) |
| **Warning table** | Top-5 certificates expiring within warning threshold (amber theme) |
| **Unknown table** | All certificates whose expiry could not be determined |
| **Legend + footer** | Status colour legend and automated disclaimer |

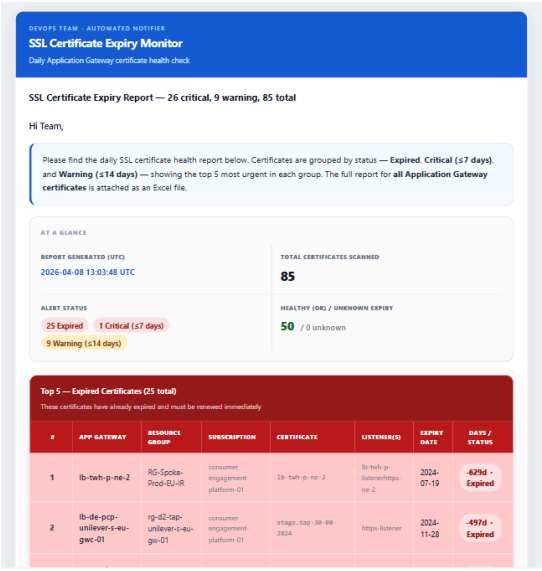
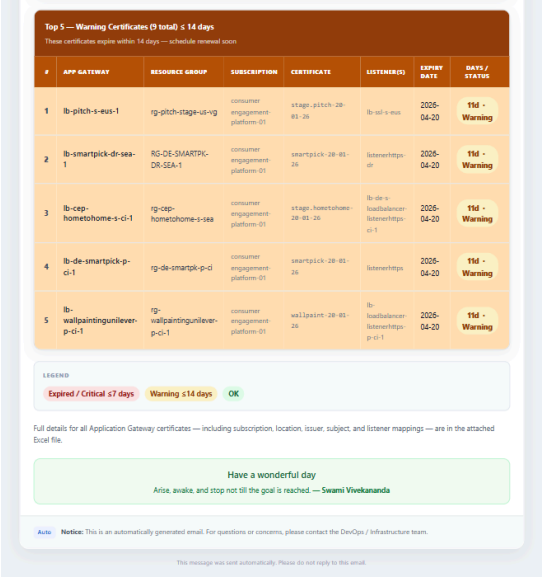



*Figure 1 — Sample HTML Email Report: colour-coded status table with metrics summary card*



Top 5 — Warning Certificates (9 total) ≤ 14 days

These certificates expire within 14 days — schedule renewal soon

| # | APP GATEWAY | RESOURCE GROUP | SUBSCRIPTION | CERTIFICATE | LISTENER(S) | EXPIRY DATE | DAYS / STATUS |
|---|---|---|---|---|---|---|---|
| 1 | lb-pitch-s-eus-1 | rg-pitch-stage-us-vg | consumer engagement-platform-01 | stage.pitch-20-01-26 | lb-ssl-s-eus | 2026-04-20 | 11d - Warning |
| 2 | lb-smartpick-dr-sea-1 | RG-DE-SMARTPK-DR-SEA-1 | consumer engagement-platform-01 | smartpick-20-01-26 | listenerhttps-dr | 2026-04-20 | 11d - Warning |
| 3 | lb-cep-hometohome-s-ci-1 | rg-cep-hometohome-s-sea | consumer engagement-platform-01 | stage.hometohome-20-01-26 | lb-de-s-loadbalancer-listenerhttps-ci-1 | 2026-04-20 | 11d - Warning |
| 4 | lb-de-smartpick-p-ci-1 | rg-de-smartpk-p-ci | consumer engagement-platform-01 | smartpick-20-01-26 | listenerhttps | 2026-04-20 | 11d - Warning |
| 5 | lb-wallpaintingunilever-p-ci-1 | rg-wallpaintingunilever-p-ci-1 | consumer engagement-platform-01 | wallpaint-20-01-26 | lb-loadbalancer-listenerhttps-p-ci-1 | 2026-04-20 | 11d - Warning |

LEGEND

Expired / Critical ≤7 days  Warning ≤14 days  OK

Full details for all Application Gateway certificates — including subscription, location, issuer, subject, and listener mappings — are in the attached Excel file.

Have a wonderful day

Arise, awake, and stop not till the goal is reached. — Swami Vivekananda

Auto **Notice:** This is an automatically generated email. For questions or concerns, please contact the DevOps / Infrastructure team.

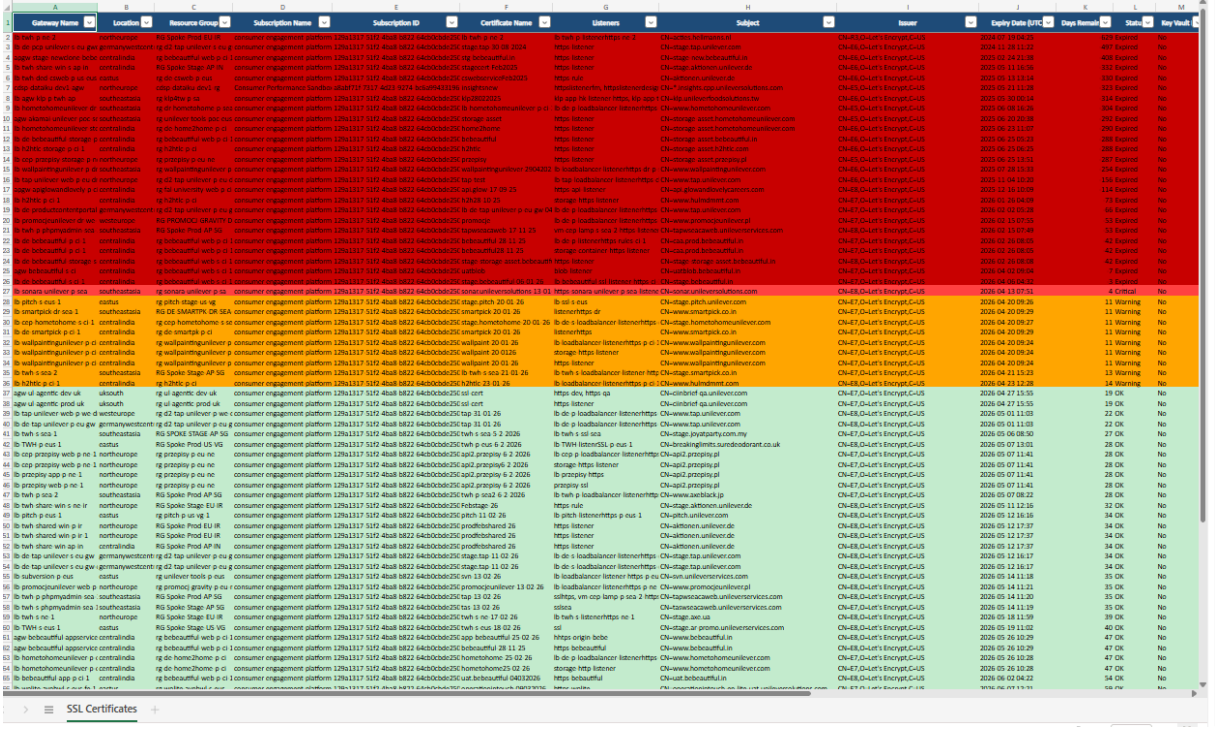
This message was sent automatically. Please do not reply to this email.

*Figure 2 — Sample HTML Email Report: colour-coded status table with metrics summary card*

## Excel Attachment Columns

| # | Column | Description |
|---|---|---|
| 1 | Gateway Name | Name of the Application Gateway |
| 2 | Location | Azure region |
| 3 | Resource Group | Resource group containing the gateway |
| 4 | Subscription Name | Resolved subscription display name |
| 5 | Subscription ID | Azure subscription ID |
| 6 | Certificate Name | Name of the SSL certificate resource |
| 7 | Listeners | TLS listeners using this certificate |
| 8 | Subject | Certificate subject (CN) |
| 9 | Issuer | Certificate issuer / CA |
| 10 | Expiry Date (UTC) | Certificate expiry date |
| 11 | Days Remaining | Calendar days until expiry |
| 12 | Status | Expired / Critical / Warning / OK / Unknown |
| 13 | Key Vault Ref | Key Vault URI if certificate is sourced from Key Vault |

All rows are sorted by days remaining (soonest-expiring first). Each row is colour-coded by status: dark red (Expired), orange-red (Critical), amber (Warning), green (OK), grey (Unknown).



| Gateway Name | Location | Resource Group | Subscription Name | Subscription ID | Certificate Name | Listeners | Subject | Issuer | Expiry Date (UTC) | Days Remain | Status | Key Vault |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| lb-hah-p-ne-2 | northeurope | RG-Spoke-Prod-EU-IR | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | lb-hah-p-ne-2 | b-hah-p-listenerhttps-ne-2 | CN=azfdm.hollmanns.nl | CN=R3,O=Let's Encrypt,C=US | 2024-07-19 04:25 | 129 | Expired | No |
| lb-de-pcp-unilever-s-eu-gw | germanywestcent | rg-d2-tap-unilever-s-eu-g | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stage-tap-30-08-2024 | https-listener | CN=stage.tap.unilever.com | CN=E6,O=Let's Encrypt,C=US | 2024-11-28 11:22 | 497 | Expired | No |
| sggw-stage-newstone-bebe | centralindia | rg-bebeautiful-web-p-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stg-bebeautiful-in | https-listener | CN=stage-new.bebeautiful.in | CN=E6,O=Let's Encrypt,C=US | 2025-02-24 21:38 | 408 | Expired | No |
| lb-twh-share-win-s-se-in | centralindia | RG-Spoke-Stage-AP-IN | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stagecert-Feb2025 | https-listener | CN=stage-aktionen.unilever.de | CN=E6,O=Let's Encrypt,C=US | 2025-05-11 16:56 | 332 | Expired | No |
| lb-twh-dod-cswsb-p-us-cus | eastus | rg-de-cswsb-p-eus | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | cswebservicefeb2025 | https-rule | CN=aktionen.unilever.de | CN=E6,O=Let's Encrypt,C=US | 2025-05-13 13:14 | 330 | Expired | No |
| cdcp-dataku-dev1-sgw | northeurope | cdcp-dataku-dev1-rg | Consumer Performance Sandbo | a8abf71f-7317-4623-9274-bc6c99433196 | insightsnew | https-listenerhttp, https-listenercdcp | CN=*.insights.cpp.unileversolutions.com | CN=R3,O=Let's Encrypt,C=US | 2025-05-21 11:28 | 323 | Expired | No |
| lb-agw-kip-p-twh-ap | southeastasia | rg-kip4tw-p-sea | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | kip28022025 | kip-app-tls-listener-https, kip-app-t | CN=kip.unileverfoodsolutions.tw | CN=E6,O=Let's Encrypt,C=US | 2025-05-30 00:14 | 314 | Expired | No |
| lb-homethohomeunilever-dr | southeastasia | rg-de-homethohome-p-sea | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | lb-homethohomeunilever-p-cl | lb-de-p-loadbalancer-listenerhttps | CN=www.homethohomeunilever.com | CN=E5,O=Let's Encrypt,C=US | 2025-06-08 16:26 | 304 | Expired | No |
| agw-akamai-unilever-poc-e | southeastasia | rg-unilever-tools-poc-eus | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | storage-asset | https-listener | CN=storage-asset.homethohomeunilever.com | CN=E5,O=Let's Encrypt,C=US | 2025-06-20 20:38 | 292 | Expired | No |
| lb-homethohomeunilever-st | centralindia | rg-de-home2home-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | home2home | https-listener | CN=storage-asset.homethohomeunilever.com | CN=E6,O=Let's Encrypt,C=US | 2025-06-23 11:07 | 290 | Expired | No |
| lb-de-bebeautiful-storage-p | centralindia | rg-bebeautiful-web-p-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | bebeautiful | https-listener | CN=storage-asset.bebeautiful.in | CN=E6,O=Let's Encrypt,C=US | 2025-06-25 03:23 | 288 | Expired | No |
| lb-h2htic-storage-p-ci-1 | centralindia | rg-h2htic-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | h2htic | https-listener | CN=storage-asset.h2htic.com | CN=E6,O=Let's Encrypt,C=US | 2025-06-25 06:25 | 288 | Expired | No |
| lb-pcp-przepisy-storage-p-n | northeurope | rg-przepisy-p-eu-ne | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | przepisy | https-listener | CN=storage-asset.przepisy.pl | CN=E5,O=Let's Encrypt,C=US | 2025-06-25 13:51 | 287 | Expired | No |
| lb-wallpaintingunilever-p-de | southeastasia | rg-wallpaintingunilever-p | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | wallpaintingunilever-2904202 | lb-loadbalancer-listenerhttps-dr-p-c | CN=www.wallpaintingunilever.com | CN=E6,O=Let's Encrypt,C=US | 2025-07-28 15:33 | 254 | Expired | No |
| lb-tap-unilever-web-p-eu-cl | northeurope | rg-d2-tap-unilever-p-eu-c | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | tap-test | lb-tap-loadbalancer-listenerhttps-c | CN=www.tap.unilever.com | CN=E6,O=Let's Encrypt,C=US | 2025-11-04 10:20 | 156 | Expired | No |
| sggw-sogiowandlovely-p-ci | centralindia | rg-fai-university-web-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | api.glow-17-09-25 | https-api-listener | CN=api.glowandlovelycancers.com | CN=E8,O=Let's Encrypt,C=US | 2025-12-16 10:09 | 114 | Expired | No |
| lb-h2htic-p-ci-1 | centralindia | rg-h2htic-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | h2h28-10-25 | storage-https-listener | CN=www.hulmdmmt.com | CN=E7,O=Let's Encrypt,C=US | 2026-01-26 04:09 | 73 | Expired | No |
| lb-de-productcontentportal | germanywestcent | rg-d2-tap-unilever-p-eu-g | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | lb-de-tap-unilever-p-eu-gw-04 | lb-de-p-loadbalancer-listenerhttps | CN=www.tap.unilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-02-02 03:28 | 66 | Expired | No |
| lb-promocjounilever-dr-we | westeurope | RG-PROMOCJ-GRAVITY-D | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | promocjo | lb-de-p-loadbalancer-listenerhttps | CN=www.promocjounilever.pl | CN=E7,O=Let's Encrypt,C=US | 2026-02-15 07:55 | 53 | Expired | No |
| lb-twh-p-phpmyadmin-sea | southeastasia | RG-Spoke-Prod-AP-SG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | tapwsecaceweb-17-11-25 | vm-cop-lamp-s-sea-2-https-listene | CN=tapwsecaceweb.unileverservices.com | CN=E8,O=Let's Encrypt,C=US | 2026-02-15 07:49 | 53 | Expired | No |
| lb-de-bebeautiful-p-ci-1 | centralindia | rg-bebeautiful-web-p-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | bebeautiful-28-11-25 | lb-de-p-listenerhttps-rules-ci-1 | CN=cas.prod.bebeautiful.in | CN=E7,O=Let's Encrypt,C=US | 2026-02-26 08:05 | 42 | Expired | No |
| lb-de-bebeautiful-p-ci-1 | centralindia | rg-bebeautiful-web-p-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | bdbeautiful28-11-25 | storage-container-https-listener | CN=cas.prod.bebeautiful.in | CN=E7,O=Let's Encrypt,C=US | 2026-02-26 08:05 | 42 | Expired | No |
| lb-de-bebeautiful-storage-s | centralindia | rg-bebeautiful-web-s-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stage-storage-asset-bebeauth | https-listener | CN=stage-storage-asset.bebeautiful.in | CN=E8,O=Let's Encrypt,C=US | 2026-02-26 08:08 | 42 | Expired | No |
| agw-bebeautiful-s-ci | centralindia | rg-bebeautiful-web-s-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | uatblob | blob-listener | CN=uatblob.bebeautiful.in | CN=E7,O=Let's Encrypt,C=US | 2026-04-02 09:04 | 7 | Expired | No |
| lb-de-bebeautiful-s-ci-1 | centralindia | rg-bebeautiful-web-s-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stagebebeauti-06-01-26 | lb-bebeautiful-uat-listener-https-c | CN=stage.bebeauti.in | CN=E7,O=Let's Encrypt,C=US | 2026-04-06 04:02 | 9 | Expired | No |
| lb-sonara-unilever-p-sea | southeastasia | rg-sonara-unilever-p-sea | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | sonar.unileversolutions-13-01 | https-sonara-unilever-p-sea-listene | CN=sonar.unileversolutions.com | CN=E8,O=Let's Encrypt,C=US | 2026-04-13 07:51 | 4 | Critical | No |
| lb-pitch-s-eus-1 | eastus | rg-pitch-stage-us-vg | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stage.pitch-20-01-26 | lb-ssl-s-eus | CN=stage.pitch.unilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-04-20 09:26 | 11 | Warning | No |
| lb-smartpick-dr-sea-1 | southeastasia | RG-DE-SMARTPK-DR-SEA | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | smartpick-20-01-26 | listenerhttps-dr | CN=www.smartpick.co.in | CN=E7,O=Let's Encrypt,C=US | 2026-04-20 09:29 | 11 | Warning | No |
| lb-ccp-hometohome-s-ci-1 | centralindia | rg-ccp-hometohome-s-se | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stage.hometohome-20-01-26 | lb-de-s-loadbalancer-listenerhttps | CN=stage.hometohomeunilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-04-20 09:27 | 11 | Warning | No |
| lb-de-smartpick-p-ci-1 | centralindia | rg-de-smartpk-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | smartpick-20-01-26 | listenerhttps | CN=www.smartpick.co.in | CN=E7,O=Let's Encrypt,C=US | 2026-04-20 09:29 | 11 | Warning | No |
| lb-wallpaintingunilever-p-ci | centralindia | rg-wallpaintingunilever-p | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | wallpaint-20-01-26 | lb-loadbalancer-listenerhttps-p-ci-1 | CN=www.wallpaintingunilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-04-20 09:24 | 11 | Warning | No |
| lb-wallpaintingunilever-p-ci | centralindia | rg-wallpaintingunilever-p | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | wallpaint-20-0126 | storage-https-listener | CN=www.wallpaintingunilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-04-20 09:24 | 11 | Warning | No |
| lb-wallpaintingunilever-p-ci | centralindia | rg-wallpaintingunilever-p | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | wallpaint-20-01-26 | https-listener | CN=www.wallpaintingunilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-04-20 09:24 | 11 | Warning | No |
| lb-twh-s-sea-2 | southeastasia | RG-Spoke-Stage-AP-SG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | lb-twh-s-sea-21-01-26 | lb-twh-s-loadbalancer-listener-https | CN=stage.smartpick.co.in | CN=E7,O=Let's Encrypt,C=US | 2026-04-21 15:23 | 13 | Warning | No |
| lb-h2htic-p-ci-1 | centralindia | rg-h2htic-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | h2htic-23-01-26 | lb-loadbalancer-listenerhttps-p-ci-1 | CN=www.hulmdmmt.com | CN=E8,O=Let's Encrypt,C=US | 2026-04-23 12:28 | 14 | Warning | No |
| agw-ul-agentic-dev-uk | uksouth | rg-ul-agentic-dev-uk | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | ssl-cert | https-dev, https-qa | CN=cinbrief-qa.unilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-04-27 15:55 | 19 | OK | No |
| agw-ul-agentic-prod-uk | uksouth | rg-ul-agentic-prod-uk | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | ssl-cert | https-listener | CN=cinbrief.qa.unilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-04-27 15:55 | 19 | OK | No |
| lb-tap-unilever-web-p-we-d | westeurope | rg-d2-tap-unilever-p-we-c | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | tap-31-01-26 | lb-de-p-loadbalancer-listenerhttps | CN=www.tap.unilever.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-01 11:03 | 22 | OK | No |
| lb-de-tap-unilever-p-eu-gw | germanywestcent | rg-d2-tap-unilever-p-eu-g | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | tap-31-01-26 | lb-de-p-loadbalancer-listenerhttps | CN=www.tap.unilever.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-01 11:03 | 22 | OK | No |
| lb-twh-s-sea-1 | southeastasia | RG-SPOKE-STAGE-AP-SG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | twh-s-sea-5-2-2026 | lb-twh-s-ssl-sea | CN=stage.joyatparty.com.my | CN=E7,O=Let's Encrypt,C=US | 2026-05-06 08:50 | 27 | OK | No |
| lb-TWH-p-eus-1 | eastus | RG-Spoke-Prod-US-VG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | twh-p-eus-6-2-2026 | lb-TWH-listenerSSL-p-eus-1 | CN=breakinglimits.suredeodorant.co.uk | CN=E8,O=Let's Encrypt,C=US | 2026-05-07 13:01 | 28 | OK | No |
| lb-ccp-przepisy-web-p-ne-1 | northeurope | rg-przepisy-p-eu-ne | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | api2.przepisy-6-2-2026 | lb-ccp-p-loadbalancer-listenerhttp | CN=api2.przepisy.pl | CN=E7,O=Let's Encrypt,C=US | 2026-05-07 11:41 | 28 | OK | No |
| lb-ccp-przepisy-web-p-ne-1 | northeurope | rg-przepisy-p-eu-ne | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | api2.przepisy6-2-2026 | storage-https-listener | CN=api2.przepisy.pl | CN=E7,O=Let's Encrypt,C=US | 2026-05-07 11:41 | 28 | OK | No |
| lb-przepisy-app-p-ne-1 | northeurope | rg-przepisy-p-eu-ne | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | api2.przepisy-6-2-2026 | lb-przepisy-https | CN=api2.przepisy.pl | CN=E7,O=Let's Encrypt,C=US | 2026-05-07 11:41 | 28 | OK | No |
| lb-przepisy-web-p-ne-1 | northeurope | rg-przepisy-p-eu-ne | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | api2.przepisy-6-2-2026 | przepisy-ssl | CN=api2.przepisy.pl | CN=E7,O=Let's Encrypt,C=US | 2026-05-07 11:41 | 28 | OK | No |
| lb-twh-p-sea-2 | southeastasia | RG-Spoke-Prod-AP-SG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | twh-p-sea2-6-2-2026 | lb-twh-p-loadbalancer-listenerhttp | CN=www.axeblack.jp | CN=E7,O=Let's Encrypt,C=US | 2026-05-07 08:22 | 28 | OK | No |
| lb-twh-share-win-s-ne-ir | northeurope | RG-Spoke-Stage-EU-IR | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | Febstage-26 | https-rule | CN=stage.aktionen.unilever.de | CN=E8,O=Let's Encrypt,C=US | 2026-05-11 12:16 | 32 | OK | No |
| lb-pitch-p-eus-1 | eastus | rg-pitch-p-us-vg-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | pitch-11-02-26 | lb-pitch-listenerhttps-p-eus-1 | CN=pitch.unilever.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-12 16:16 | 34 | OK | No |
| lb-twh-shared-win-p-ir | northeurope | RG-Spoke-Prod-EU-IR | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | prodfebshared-26 | https-listener | CN=aktionen.unilever.de | CN=E8,O=Let's Encrypt,C=US | 2026-05-12 17:37 | 34 | OK | No |
| lb-twh-shared-win-p-ir-1 | northeurope | RG-Spoke-Prod-EU-IR | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | prodfebshared-26 | https-listener | CN=aktionen.unilever.de | CN=E8,O=Let's Encrypt,C=US | 2026-05-12 17:37 | 34 | OK | No |
| lb-twh-share-win-ap-in | centralindia | RG-Spoke-Prod-AP-IN | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | prodfebshared-26 | https-listener | CN=aktionen.unilever.de | CN=E8,O=Let's Encrypt,C=US | 2026-05-12 17:37 | 34 | OK | No |
| lb-de-tap-unilever-s-eu-gw | germanywestcent | rg-d2-tap-unilever-p-eu-g | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stage.tap-11-02-26 | lb-de-s-loadbalancer-listenerhttps | CN=stage.tap.unilever.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-12 16:17 | 34 | OK | No |
| lb-de-tap-unilever-s-eu-gw | germanywestcent | rg-d2-tap-unilever-p-eu-g | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | stage.tap-11-02-26 | lb-de-s-loadbalancer-listenerhttps | CN=stage.tap.unilever.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-12 16:17 | 34 | OK | No |
| lb-subversion-p-eus | eastus | rg-unilever-tools-p-eus | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | svn-13-02-26 | lb-loadbalancer-listener-https-p-eu | CN=svn.unileverservices.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-14 11:18 | 35 | OK | No |
| lb-promocjounilever-web-p | northeurope | rg-promocj-gravity-p-eu-c | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | promocjounilever-13-02-26 | lb-loadbalancer-listenerhttps-p-ne | CN=www.promocjounilever.pl | CN=E8,O=Let's Encrypt,C=US | 2026-05-14 11:21 | 35 | OK | No |
| lb-twh-p-phpmyadmin-sea | southeastasia | RG-Spoke-Prod-AP-SG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | tap-13-02-26 | sslhttps, vm-ccp-lamp-p-sea-2-http | CN=tapwsecaceweb.unileverservices.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-14 11:20 | 35 | OK | No |
| lb-twh-s-phpmyadmin-sea-1 | southeastasia | RG-Spoke-Stage-AP-SG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | tas-13-02-26 | sslsea | CN=tawsecaceweb.unileverservices.com | CN=E7,O=Let's Encrypt,C=US | 2026-05-14 11:19 | 35 | OK | No |
| lb-twh-s-ne-1 | northeurope | RG-Spoke-Stage-EU-IR | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | twh-s-ne-17-02-26 | lb-twh-s-listenerhttps-ne-1 | CN=stage.axe.ua | CN=E8,O=Let's Encrypt,C=US | 2026-05-18 11:39 | 39 | OK | No |
| lb-TWH-s-eus-1 | eastus | RG-Spoke-Stage-US-VG | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | twh-s-eus-18-02-26 | ssl | CN=stage.ar.promo.unileverservices.com | CN=E8,O=Let's Encrypt,C=US | 2026-05-19 11:02 | 40 | OK | No |
| agw-bebeautiful-appservice | centralindia | rg-bebeautiful-web-p-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | app-bebeautiful-25-02-26 | https-origin-bebe | CN=www.bebeautiful.in | CN=E8,O=Let's Encrypt,C=US | 2026-05-26 10:29 | 47 | OK | No |
| agw-bebeautiful-appservice | centralindia | rg-bebeautiful-web-p-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | bebeautiful-28-11-25 | https-bebeautiful | CN=www.bebeautiful.in | CN=E8,O=Let's Encrypt,C=US | 2026-05-26 10:29 | 47 | OK | No |
| lb-hometohomeunilever-p-c | centralindia | rg-de-home2home-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | hometohome-25-02-26 | lb-de-p-loadbalancer-listenerhttps | CN=www.hometohomeunilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-05-26 10:28 | 47 | OK | No |
| lb-hometohomeunilever-p-c | centralindia | rg-de-home2home-p-ci | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | hometohome25-02-26 | storage-http-listener | CN=www.hometohomeunilever.com | CN=E7,O=Let's Encrypt,C=US | 2026-05-26 10:28 | 47 | OK | No |
| lb-bebeautiful-app-p-ci-1 | centralindia | rg-bebeautiful-web-p-ci-1 | consumer-engagement-platform | 129a1317-51f2-4ba8-b822-64c0cbdc25c | uat.bebeautiful-04032026 | https-bebeautiful | CN=uat.bebeautiful.in | CN=E8,O=Let's Encrypt,C=US | 2026-06-02 04:22 | 54 | OK | No |

SSL Certificates

*Figure 3 — sample excel report*

## 14 Advantages

### Proactive Monitoring

- **Daily schedule** — catches certificates before they expire, not after outages occur
- **Configurable warning threshold** — adjust lead time to match your team's renewal process
- **Multi-subscription** — single runbook covers the entire Azure estate from one place

### Safety

- **Read-only** — zero risk of accidental resource modification under any circumstances
- **Minimal RBAC** — Reader role only; no elevated or write permissions required
- **Dry-run mode** — full local testing without any email or Azure API side effects

### Resilience

- **4-stage certificate parsing** — handles all certificate formats Azure may return across different gateway versions and configurations
- **Listener-aware** — only scans certificates actively attached to TLS listeners
- **Unknown classification** — explicitly handles Key Vault-backed certificates rather than silently failing or crashing

### Reporting Quality

- **Dual output** — HTML email for at-a-glance visibility, Excel for detailed audit records
- **Colour-coded** — immediate visual prioritisation by severity status
- **Sorted by urgency** — soonest-expiring certificates always appear first in both outputs
- **Compliance-ready** — Excel output suitable as audit evidence for SOC 2, ISO 27001, and similar frameworks

## 15 Limitations

| Limitation | Impact |
|---|---|
| **Key Vault-backed certificates** | Azure does not expose `publicCertData` for Key Vault certs — these appear as "Unknown" and must be monitored via Azure Key Vault expiry alerts |
| **No auto-renewal** | Notification only — the runbook does not renew, replace, or rotate any certificate |
| **Python 3.8 runtime** | Must remain compatible with the Azure Automation Python 3.8 runtime; some newer library features may require version pinning |
| **No historical trend** | Each run is independent; there is no built-in storage of past results or trend data |
| **Application Gateway only** | Does not scan certificates on Front Door, API Management, App Service, or other Azure services |

| Limitation | Impact |
| --- | --- |
| **ACS Email single-region** | The ACS connection string is tied to a specific ACS resource region |
| **Top-5 per status in email** | HTML email shows only the top-5 most urgent per status to keep email readable; full data is in the Excel attachment |

## 16 Use Cases

- **Enterprise SSL governance** — enforce certificate visibility across all Application Gateways in an Azure estate from a single automated runbook
- **Proactive incident prevention** — eliminate SSL-related outages caused by unnoticed certificate expiry across multiple subscriptions
- **Compliance requirements** — supports SOC 2, ISO 27001, and other frameworks requiring certificate lifecycle monitoring and audit evidence
- **Multi-subscription Azure environments** — centralised visibility across dev, staging, and production subscriptions without per-subscription tooling
- **Operations team dashboards** — daily email summary gives ops teams immediate visibility without portal access

# 17 Operational Runbook

## Trigger the Runbook Manually

Navigate to the Automation Account in the Azure Portal → **Runbooks** → select the runbook → **Start**.

## Test Locally

```
pip install -r requirements.txt
DRY_RUN=true python main.py   # writes .xlsx + .html to disk, no email sent
```

## Change the Warning Threshold

Update the `AG_ALERT_DAYS` variable in the Automation Account. No runbook redeployment is required.

## Add a New Subscription

1. Add the subscription ID to the `AG_SUBSCRIPTION_IDS` Automation variable (comma-separated)
2. Grant the Service Principal `Reader` role on the new subscription

## Add or Change Email Recipients

Update the `AG_NOTIFICATION_TO` Automation variable. No runbook redeployment is required.

## Check Execution History

Navigate to the Automation Account → **Jobs** to view past executions, output logs, and error details.

## Common Issues

| Symptom | Likely Cause | Resolution |
|---|---|---|
| All certificates show as Unknown | Certificates are Key Vault-backed; `publicCertData` not available | Monitor Key Vault-backed certs via Azure Key Vault expiry alerts |
| Authentication error at startup | `AzureSPCredential` missing or incorrect in Automation Account | Re-add the credential with correct Client ID and Secret |
| No email received | ACS connection string or sender address incorrect | Verify `ACS_CONNECTION_STRING` and `ACS_SENDER_ADDRESS` variables; check sender domain verification in ACS |
| Package import errors | Required Python packages not installed in Automation Account | Upload all packages from `requirements.txt` to the Automation Account Python 3 packages |

## 18 Final Outcome

**This solution delivers a production-grade, enterprise-ready SSL certificate monitoring system**

- **Proactive expiry monitoring** — daily scheduled scan catches expiring certificates before they cause outages or impact end users
- **Multi-subscription coverage** — single runbook covers all Application Gateways across the entire Azure estate
- **Resilient certificate parsing** — four-stage fallback chain handles every certificate format Azure returns across all gateway versions
- **Enterprise-grade reporting** — colour-coded HTML email + full Excel attachment delivered to stakeholders after every run
- **Read-only safety** — zero risk of resource modification; minimal `Reader` RBAC permissions required
- **Compliance-ready** — daily Excel audit trail suitable for SOC 2, ISO 27001, and similar governance frameworks